

HW 06 — 1D Lagrangian Hydrodynamics: Sod Shock Tube

Overview

This assignment implements a staggered Lagrangian finite-difference scheme to solve the 1D Euler equations of compressible gas dynamics. The test problem is the **Sod shock tube**: a membrane separating two regions of different pressure is removed at $t = 0$, and three wave structures emerge — a rarefaction fan, a contact discontinuity, and a shock. The simulation is run on four grid resolutions ($N = 34, 100, 334, 1000$) and compared against the analytic exact solution.

Problem Statement

Evolve the ideal-gas Euler equations in 1D on the domain $[0, 16]$ from $t = 0$ to $t = 2.5$, with initial condition:

$$(\rho, u, p) = \begin{cases} (1.0, 0, 1.0) & x < 8 \\ (0.125, 0, 0.1) & x > 8 \end{cases}$$

Rigid walls at both ends. Equation of state: $p = (\gamma - 1)\rho e$ with $\gamma = 1.4$.

Compare numerically measured feature speeds (shock, contact, rarefaction head and tail) against the exact Riemann solution, and check energy conservation.

The Physical Waves (Exact Solution)

The Sod problem produces three waves traveling at speeds determined by the Rankine-Hugoniot and isentropic relations:

Feature	Speed (exact)	Description
Rarefaction head	-1.183	Left edge of the fan; moves left at $-c_L$
Rarefaction tail	-0.070	Right edge of the fan; barely moves
Contact discontinuity	$+0.927$	Jump in ρ only; no pressure or velocity jump
Shock	$+1.752$	Compression wave; jumps in ρ, p, u all

The intermediate pressure p_* is the root of $f_L(p_*) + f_R(p_*) + (u_R - u_L) = 0$, where each f_K is a branch (rarefaction or shock) depending on whether $p_* < p_K$ or $p_* > p_K$. This is solved with `brentq`.

The Numerical Scheme

Staggered Lagrangian Grid

The grid is **staggered**: two interleaved sets of degrees of freedom move with the fluid (Lagrangian frame):

- **Cells** ($i - \frac{1}{2}$): hold ρ, e, p, q (thermodynamic quantities, cell-centered).
- **Vertices** (i): hold x, u, a (position, velocity, acceleration).

Cell mass is conserved exactly: $m_{i-1/2} = \rho_{i-1/2} V_{i-1/2}$ is fixed. When vertices move, the cell volume changes and density is recomputed from $\rho = m/V$. This automatically satisfies mass conservation to machine precision — a key advantage of the Lagrangian approach.

Initial setup:

```
x = np.linspace(X_L, X_R, N + 1)  # N+1 vertices
xc = 0.5*(x[:-1] + x[1:])          # cell centres
rho = np.where(xc < X_DIAPH, RHO_L, RHO_R)
p = np.where(xc < X_DIAPH, P_L, P_R)
e = p / ((GAMMA - 1.0) * rho)
m = rho * np.diff(x)               # fixed cell masses
```

The Time-Step Algorithm (Leapfrog / Predictor-Corrector)

Each time step advances from t^n to t^{n+1} . The scheme is a **2nd-order predictor-corrector** (sometimes called staggered leapfrog):

Step 1 — Predictor velocity (half-step):

$$u_i^{n+1/2} = u_i^n + \frac{\Delta t}{2} a_i^n$$

Walls enforced: $u_0 = u_N = 0$.

Step 2 — New positions:

$$x_i^{n+1} = x_i^n + \Delta t u_i^{n+1/2}$$

Step 3 — New volumes and densities:

$$V_{i-1/2}^{n+1} = x_i^{n+1} - x_{i-1}^{n+1}, \quad \rho_{i-1/2}^{n+1} = m_{i-1/2} / V_{i-1/2}^{n+1}$$

Step 4 — Artificial viscosity (compression only — no viscosity in expanding regions):

$$q_{i-1/2}^{n+1} = \begin{cases} \sigma \rho^{n+1} (u_{i-1}^{n+1/2} - u_i^{n+1/2})^2 & u_{i-1} > u_i \text{ (compression)} \\ 0 & \text{otherwise} \end{cases}$$

The parameter $\sigma = 3$ controls the shock-smearing width. Artificial viscosity is necessary: without it, the shock would produce unphysical oscillations (Gibbs phenomenon) in the finite-difference scheme. It converts kinetic energy into thermal energy over a few cells near the shock, mimicking the real dissipation in a thin shock layer.

Step 5 — Energy update (algebraic implicit):

The internal energy evolves by work done against pressure and viscosity:

$$e^{n+1} = \frac{e^n - \frac{\Delta V}{2m} (p^n + q^n + q^{n+1})}{1 + (\gamma - 1) \frac{\Delta V}{2V^{n+1}}}$$

This form is **algebraically implicit** in e^{n+1} and $p^{n+1} = (\gamma - 1)\rho^{n+1}e^{n+1}$ simultaneously. For an ideal gas the system decouples analytically into a single formula, avoiding an iterative solve.

Step 6 — New accelerations:

$$a_i^{n+1} = -2 \frac{(p+q)_{i+1/2}^{n+1} - (p+q)_{i-1/2}^{n+1}}{m_{i-1/2} + m_{i+1/2}}$$

Interior vertices only; walls held at $a = 0$.

Step 7 — Corrector velocity:

$$u_i^{n+1} = u_i^{n+1/2} + \frac{\Delta t}{2} a_i^{n+1}$$

Time Step Control

The time step must satisfy the **Courant-Friedrichs-Lewy (CFL) condition** — no information can travel more than one cell per step:

$$\Delta t \leq \text{CFL} \cdot \min_i \frac{\Delta x_i}{c_i}$$

where $c_i = \sqrt{\gamma p_i / \rho_i}$ is the local sound speed. Additionally, a density-change limit prevents the density from changing by more than 5% per step:

$$\Delta t \leq \frac{0.05 \Delta x_i}{|\Delta u_i|}$$

A growth factor of 1.1 prevents the step from increasing too quickly after a refined region is passed.

```
dt = CFL * np.min(dx / c)
du = np.abs(s['u'][1:] - s['u'][:-1])
dt = min(dt, MAX_DRHO * np.min(dx[moving] / du[moving]))
dt = min(dt, DT_GROWTH * dt_prev)
```

Measuring Feature Speeds

Feature speeds are measured by locating each feature on two closely spaced snapshots (at $t = 2.0$ and $t = 2.2$) and computing $v = \Delta x / \Delta t$:

- **Shock:** rightmost crossing of the pressure level $\frac{1}{2}(p_* + p_R)$.
- **Contact:** crossing of the density level $\frac{1}{2}(\rho_{*L} + \rho_{*R})$ between the fan and the shock.
- **Rarefaction head:** crossing of $0.995 \rho_L$ (just inside the undisturbed left region).
- **Rarefaction tail:** crossing of $1.005 \rho_{*L}$ (just inside the star-left plateau).

Linear interpolation is used to locate the crossing precisely between grid points.

Results

Energy Conservation

N	$\Delta E/E_0$
34	6.4×10^{-5}
100	2.1×10^{-5}
334	6.3×10^{-6}
1000	2.1×10^{-6}

The energy non-conservation scales as N^{-1} (first-order), consistent with the first-order artificial viscosity scheme.

Feature Speeds vs. Exact

N	v_{shock}	v_{contact}	v_{head}	v_{tail}
34	1.795	1.046	-1.293	-0.495
100	1.717	0.941	-1.145	+0.024
334	1.763	0.928	-1.182	-0.076
1000	1.753	0.927	-1.176	-0.074
exact	1.752	0.927	-1.183	-0.070

The shock, contact, and head converge to the exact values at $N = 334$ -1000. The **tail** is the hardest feature to resolve: it moves at only -0.07 cm/s and sits inside the diffuse rarefaction fan. It requires $N \geq 334$ cells to even locate correctly.

Physical Interpretation

- The **rarefaction fan** is well-captured at all resolutions because it is a smooth expansion wave — finite differences handle smooth regions well.
- The **shock and contact** are smeared over $\sim \sigma = 3$ cell widths by the artificial viscosity; they sharpen as N increases.
- The shock reaches $x \approx X_{\text{diaph}} + v_{\text{shock}} \cdot t \approx 12.4$ at $t = 2.5$, well inside the domain $[0, 16]$, so the rigid walls never interfere with the solution.

Key Takeaways

- **Lagrangian grids move with the fluid**, so cell mass is conserved *exactly* (density is just $\rho = m/V$). This is the defining advantage over Eulerian (fixed-grid) schemes.
- **Staggered variables**: thermodynamic quantities (ρ, e, p) live at cell centers; kinematic quantities (x, u, a) live at vertices. This staggering gives natural second-order-accurate centered differences.
- **A predictor-corrector (leapfrog) time step** advances velocity at half-steps and positions/energy at full steps — second-order accurate in smooth regions.
- **Shocks need artificial viscosity**. Without it, finite-difference schemes produce spurious oscillations at discontinuities; the viscosity $q \propto \rho(\Delta u)^2$ (active only in compression) spreads each shock over a few cells and converts kinetic energy to heat.

- **The CFL condition** $\Delta t \leq \text{CFL} \cdot \min(\Delta x/c)$ is the stability limit for explicit schemes: information must not cross more than one cell per step.
- **Always validate against an exact solution.** The Sod problem has an analytic Riemann solution, so measured wave speeds and energy drift can be checked quantitatively.
- **Resolution trade-offs:** smooth features (rarefaction fan) converge fast; discontinuities (shock, contact) stay smeared by viscosity and sharpen only as N grows. Energy error falls $\sim N^{-1}$ — first-order, set by the viscosity at the shock.